As we have said, the workflows on Databricks are very Databricks-specific. Teams can use the infrastructure to manage access rights for data products, and analysts can use the unity data catalog to explore data.

The data mesh principles work out a bit differently if we employ an architecture like this. The Databricks-centric architecture limits the freedom of teams quite a bit, and there is no easy way to opt out of the platform. Hence, this setup works well only if you either work around these limitations or make sure via governance measures that the teams are aligned in the tool choice. With this, this architecture does bring the risk of slipping back into a non-data-mesh-like mode of operation.

This makes this architecture much more suitable for younger companies. This architecture is also aimed more at data engineers and data scientists. This means it targets companies with a higher percentage of data people inside, whereas the GCP and AWS options are also accessible for software engineers.

8.3.5 Variations

Databricks in itself could be replaced by other bigger data platforms like the Snowflake ecosystem, which already offers almost all of the components needed to implement the preceding architecture.

To scale up this kind of data mesh architecture, you'll need to enable the opt-out option for every participant. This can be achieved by adding an external data catalog and reducing Databricks mostly to the transformation tooling, which still is a powerful toolset. Your setup could look like this:

1. Teams store data in their own data stores.
2. Ingestion tooling takes that data and puts it into Databricks-accessible data stores.
3. Databricks is used to do transformations on these datasets and to produce data products.
4. A query engine like Trino is used to connect various data products together.